

Problem Set 3

Applied Stats/Quant Methods

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Due: November 27, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 27, 2025. No late assignments will be accepted.

Question 1: Economics

In this question, use the **prestige** dataset in the **car** library. First, run the following commands:

```
install.packages(car)
library(car)
data(Prestige)
help(Prestige)
```

We would like to study whether individuals with higher levels of income have more prestigious jobs. Moreover, we would like to study whether professionals have more prestigious jobs than blue and white collar workers.

- (a) Create a new variable **professional** by recoding the variable **type** so that professionals are coded as 1, and blue and white collar workers are coded as 0 (Hint: **ifelse**).
- (b) Run a linear model with **prestige** as an outcome and **income**, **professional**, and the interaction of the two as predictors (Note: this is a continuous $\times$ dummy interaction.)
- (c) Write the prediction equation based on the result.
- (d) Interpret the coefficient for **income**.
- (e) Interpret the coefficient for **professional**.
- (f) What is the effect of a \$1,000 increase in income on prestige score for professional occupations? In other words, we are interested in the marginal effect of income when the variable **professional** takes the value of 1. Calculate the change in $\hat{y}$ associated with a \$1,000 increase in income based on your answer for (c).
- (g) What is the effect of changing one's occupations from non-professional to professional when her income is \$6,000? We are interested in the marginal effect of professional jobs when the variable **income** takes the value of 6,000. Calculate the change in $\hat{y}$ based on your answer for (c).

Answer (a)

Originally the observations in the variable `type` were coded as professionals, white collar or blue collar. In order to only make a distinction between professionals and non-professionals the `ifelse` function is used: `prof` are coded as 1, everything else as 0.

```
1 Prestige$professional <- ifelse(Prestige$type == "prof", 1, 0)
```

Answer (b)

Here follows the code for the multivariate regression having as independent variables `income`, `professional` and their interaction term. Afterwards the results of the estimation are presented.

```
1 model_econ <- lm(prestige ~ income * professional, data = Prestige)
```

Table 1: Economics multivariate regression

	<i>Dependent variable:</i>
	Prestige of one's occupation
Income level	0.003*** (0.0005)
Professional occupation	37.781*** (4.248)
income:professional	-0.002*** (0.001)
Constant	21.142*** (2.804)
Observations	98
R ²	0.787
Adjusted R ²	0.780
Residual Std. Error	8.012 (df = 94)
F Statistic	115.878*** (df = 3; 94)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Answer (c)

The prediction equation considers the above estimated coefficient to predict unknown **prestige** values.

$$\widehat{\text{prestige}}_i = 21.1422 + 0.0032 * \text{income}_i + 37.7813 * \text{professional}_i - 0.0023 * \text{income}_i * \text{professional}_i$$

Answer (d)

In this interaction term, the coefficient for **income** represents the average increase in prestige for a non-professional when considering a one-unit increase in income. Given that this is an interaction model, the specification regarding professional categorization is necessary: when one's job is non-professional, the variable takes the value 0. The estimated coefficient $\hat{\beta}_1 = 0.0032$ implies that, for each one-unit increase in income, a non-professional will observe a 0.0032 point increase in occupational prestige.

When an individual is a professional the effect of a unital increase in income takes the values of $\hat{\beta}_1 + \hat{\beta}_3$.

Answer (e)

The interpretation of the **professional** coefficient is bit more abstract: it represents the average difference in prestige between a professional and a non-professional (white- or blue-collar) job when income is equal to 0. This scenario is hardly realistic, since anyone who holds a job (whether professional or not) would typically receive some income.

Nonetheless, the baseline average difference between the two categories is 37.7813 prestige points. When income is taken into account, the difference in average prestige between the two groups is obtained by adding the coefficient $\hat{\beta}_2$ to the product of $\hat{\beta}_3$ and the income level. When an individual is a professional the effect of a unital increase in income takes the values of $\hat{\beta}_1 + \hat{\beta}_3$.

Answer (f)

The effect associated with a \$1,000 increase in income for professional can be calculated by summing the $\hat{\beta}_1$ and the $\hat{\beta}_3$ coefficients and multiplying the sum by 1000.

```
1 coef_income <- coef(model_econ) ["income"]
2 coef_interaction <- coef(model_econ) ["income:professional"]
3 sum_coef <- coef_income + coef_interaction
```

```
> as.numeric(sum_coef)*1000
[1] 0.8452
```

A \$1,000 increase in income increases the prestige of professionals by 0.8452 points.

The claim is tested in R and shown in the following section.

```
1 increase_income <- data.frame(
2   income = c(2000, 3000),
3   professional = c(1, 1)
4 )
5 # With mock data we compute the difference in the predicted prestige
6 # for professional with a 1000 dollars difference in income
7 predictions_income <- predict(model_econ, newdata = increase_income)
8 difference_income <- predictions_income[[2]] - predictions_income[[1]]
9
10 coef_income <- coef(model_econ)["income"]
11 coef_interaction <- coef(model_econ)["income:professional"]
12 sum_coef <- coef_income + coef_interaction
13
14 # The test is conducted to test wheter is TRUE that the difference
15 # and the sum of the coefficients multiplied by 1000 are equal
16 test <- isTRUE(all.equal(
17   as.numeric(difference_income),
18   as.numeric(sum_coef) * 1000
19 ))
```

```
> print(test)
[1] TRUE
```

Answer (g)

The effect on prestige related to a change in occupation (from non-professional to professional), considering an income of \$6,000, can be computed by summing $\hat{\beta}_2$ and the product of $\hat{\beta}_3$ and 6,000.

```
1 coef_professional <- coef(model_econ)["professional"]
2 coef_interaction <- coef(model_econ)["income:professional"]
3 sum_coef_2 <- coef_professional + coef_interaction * 6000
```

```
> as.numeric(sum_coef_2)
[1] 23.82703
```

At an income level of \$6,000, the change to a professional job from a non-professional one is associated with an increase of 23.82703 prestige points.

Once again, following similar procedures, the claim is tested in R and reported here.

```

1 change_professional <- data.frame(
2   income = c(6000, 6000),
3   professional = c(0, 1)
4 )
5 predictions_change <- predict(model_econ, newdata = change_professional)
6 difference_change <- predictions_change[[2]] - predictions_change[[1]]
7
8 coef_professional <- coef(model_econ)["professional"]
9 coef_interaction <- coef(model_econ)["income:professional"]
10 sum_coef_2 <- coef_professional + coef_interaction * 6000
11
12 test <- isTRUE(all.equal(
13   as.numeric(difference_change),
14   as.numeric(sum_coef_2)
15 ))

```

```

> print(test)
[1] TRUE

```

Question 2: Political Science

Researchers are interested in learning the effect of all of those yard signs on voting preferences.¹ Working with a campaign in Fairfax County, Virginia, 131 precincts were randomly divided into a treatment and control group. In 30 precincts, signs were posted around the precinct that read, “For Sale: Terry McAuliffe. Don’t Sellout Virginia on November 5.”

Below is the result of a regression with two variables and a constant. The dependent variable is the proportion of the vote that went to McAuliffe’s opponent Ken Cuccinelli. The first variable indicates whether a precinct was randomly assigned to have the sign against McAuliffe posted. The second variable indicates a precinct that was adjacent to a precinct in the treatment group (since people in those precincts might be exposed to the signs).

Impact of lawn signs on vote share	
Precinct assigned lawn signs (n=30)	0.042 (0.016)
Precinct adjacent to lawn signs (n=76)	0.042 (0.013)
Constant	0.302 (0.011)

Notes: $R^2=0.094$, N=131

- (a) Use the results from a linear regression to determine whether having these yard signs in a precinct affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).
- (b) Use the results to determine whether being next to precincts with these yard signs affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).
- (c) Interpret the coefficient for the constant term substantively.

¹Donald P. Green, Jonathan S. Krasno, Alexander Coppock, Benjamin D. Farrer, Brandon Lenoir, Joshua N. Zingher. 2016. “The effects of lawn signs on vote outcomes: Results from four randomized field experiments.” *Electoral Studies* 41: 143-150.

- (d) Evaluate the model fit for this regression. What does this tell us about the importance of yard signs versus other factors that are not modeled?

Answer (a)

In verifying whether having the yard signs in a precinct affects vote share we conduct an hypothesis test on the related $\hat{\beta}$. In our analysis:

1. H_0 states: $\hat{\beta}_{\text{treatment}} = 0$, the effect on Cuccinelli's vote share of having the signs is equal to 0.
2. H_a states: $\hat{\beta}_{\text{treatment}} \neq 0$, the yard signs have a significant effect on vote share.

To test this hypothesis, we compute the t-statistic associated with the estimated $\hat{\beta}_{\text{treatment}}$ coefficient. This is obtained by subtracting the null value (0) from the estimate and dividing the result by its standard error. Afterwards, we compute the associated p-value using a Student-t's distribution with $n-k-1$ degrees of freedom.

```
1 t_stat <- 0.042 / 0.016
2 df <- (131 - 2 - 1) # 128
3 p_value <- 2 * pt(abs(t_stat), df = df, lower.tail = FALSE)
4 alpha <- 0.05
5 result <- ifelse(p_value > alpha,
6                 paste("p-value:", signif(p_value, 4), "- Ho cannot be
    rejected"),
```

```
> print(result)
[1] "p-value: 0.00972001974200814 - Ho is rejected"
```

It's found that having yard signs has a statistically significant effect on the vote share of the candidate Cuccinelli. This effect is statistically significant at the 95% confidence level.

Answer (b)

Similar procedures as the ones conducted above will be carried out to verify whether the effect of being next to precincts with assigned yard signs on vote shares is significant. The hypothesis tested is the following:

1. H_0 states: $\hat{\beta}_{\text{adjacent.treat}} = 0$, meaning that the effect on Cuccinelli's vote share of a precinct being adjacent to one assigned to have yard signs is zero.
2. H_a states: $\hat{\beta}_{\text{adjacent.treat}} \neq 0$, the closeness to a precinct having yard signs have a significant effect on vote share.

As anticipated, the procedures to test the hypothesis is the same as above.

```

1 t_stat_b <- 0.042 / 0.013
2 df <- (131 - 2 - 1)
3 p_value_b <- 2 * pt(abs(t_stat_b), df = df, lower.tail = FALSE)
4 result_b <- ifelse(p_value_b > alpha,
5                   paste("p-value:", signif(p_value_b, 4), "- Ho cannot be
6                   rejected"),
7                   paste("p-value:", signif(p_value_b, 4), "- Ho is rejected"
8                   ))

```

```

> print(result_b)
[1] "p-value: 0.001569 - Ho is rejected"

```

The results show that having an adjacent precinct with yard signs against McAuliffe has, on average, a positive effect of Cuccinelli's vote share. The results hold at a 5% significant level.

Answer (c)

The constant term is interpretable as the value of y when both X_1 and X_2 are equal to 0. In the case under examination $\hat{\beta}_0 = 0.302$ would be Cuccinelli's vote share when the precinct doesn't have yard signs against McAuliffe nor it is adjacent to a precinct having them. In this scenario Cuccinelli's vote share would be equal to 30.2%.

Answer (d)

To test whether the estimated coefficients effect vote share we can conduct an F-test. The procedure would help verify whether both the coefficients related to the presence of lawn signs are equal to 0, it does so by considering whether the R^2 value of the full model is significantly higher than that of the sole constant.

```

1 # df_2 are the added variable in the full model when compared to the base
  # model
2 # (i.e. the model with just the constant)
3 df_2 <- 2
4 f_value <- (0.094 / df_2) / ((1-0.094)/(df))
5 p_value_F <- pf(f_value, df1 = df_2, df2 = df, lower.tail = FALSE)

```

```
> print(p_value_F)
[1] 0.001803868
```

The results are in line with the individual t-test conducted above. A p-value equal to 0.0018 makes us reject the null hypothesis of both $\hat{\beta}_{\text{treatment}}$ and $\hat{\beta}_{\text{adjacent_treat}}$ are equal to.

While the estimated coefficients are statistically significant, the model using only these predictors explains little of the variance in vote share. The R^2 being equal to 0.094 implies only 9.4% of the variance observed in Cuccinelli's vote share is explained by the presence of lawn signs.

Moreover, this value suggests that other factors not included in the model should be more deeply investigated and potentially incorporated. Some examples of those other explanatory variables might be party ID, sex, age and income levels.